INTR – Accepted if IF=1
Ignored IF=0

4

- $\overline{RD}$ (Read) — 0 : Enabled; 1 : Disabled
- $\overline{WR}$ (Write) — 0: Enabled; 1: Disabled
- $M/\overline{IO}$ (Memory or I/O) — 0: I/O is selected; 1: Mem is "
- Ready — 0: Not ready; 1: Ready
- RESET: If this pin is kept high for 4 consecutive clock periods the 8086 gets reset.

40 pin DIP

✓ 16 data lines
✓ 19 add. lines
· Multiplexed

$\overline{BHE}$ (Bus High Enable)
0 (HMB Enabled)
1 (" disabled)

1 0

t_c

sinusoidal

$t_c = \frac{1}{5\times10^6}$ sec

$\frac{1}{2\times5\times10^6}$ sec

t_c t_c

1

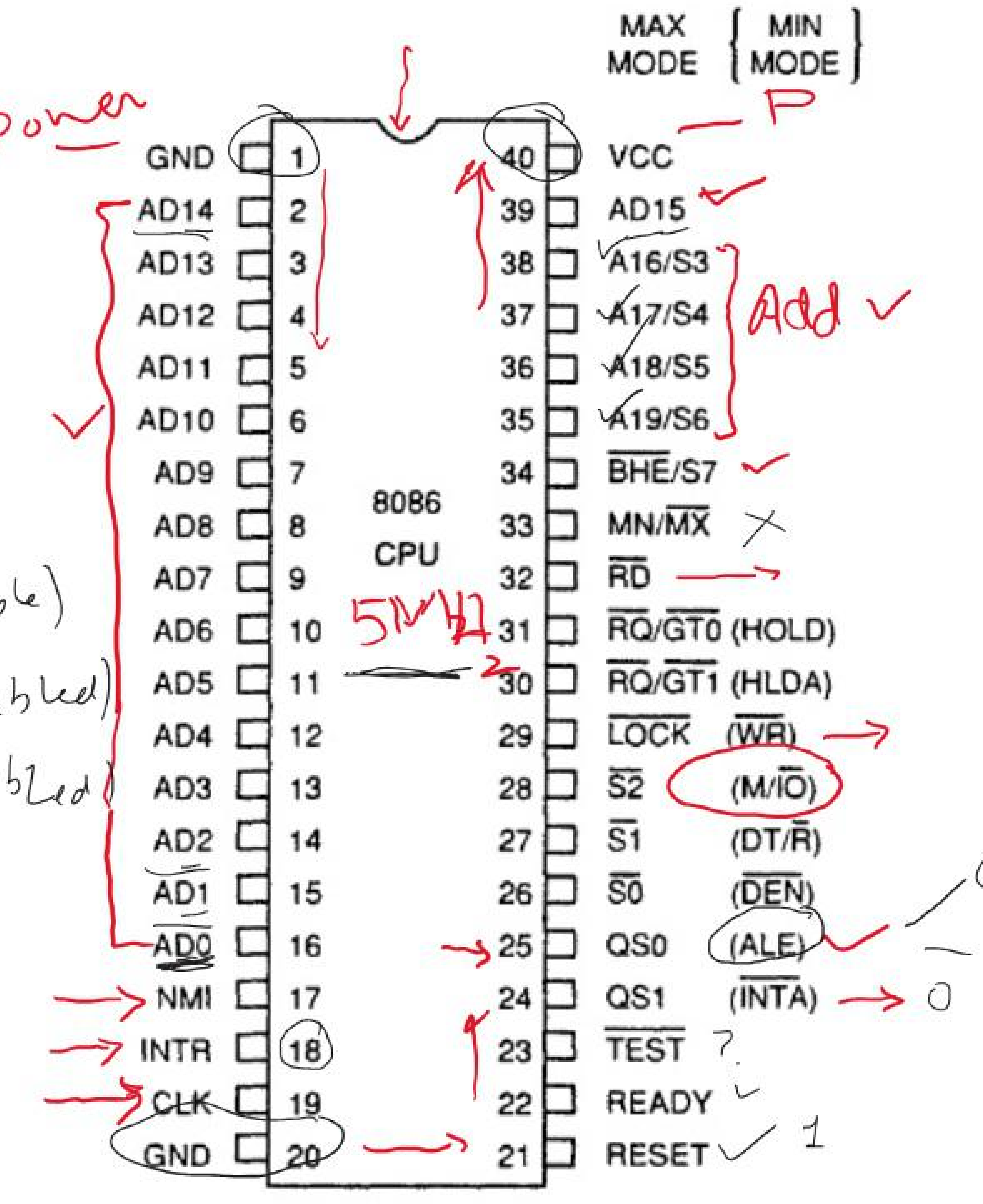
0

power

P

Add

5MHz



(a)

0 - Data
1 - Address

0

1

Non-Maskable Interrupt
Interrupts are accepted regardless the IF

-2-

Logic Level	Voltage	Current
0 (0 ~ 0.8v)	0.8 V maximum	±10 µA maximum
1 (2 ~ 5.0v)	2.0 V minimum	±10 µA maximum

1.5v ?

$M/\overline{IO}$

1 / 0

Non-deterministic

High-impedence

μ-P 8086 (1MB)

$A_{19} \sim A_0$

19

18

a. 512kB

b. 256kB

c. 128 kB

Mem μP

$MA \leq \mu A$

$MO = \mu D$

$A_{19} . A_{18}$

$D_{15} \sim D_0$

$\overline{RD}$ 1

$\overline{WR}$ 0

? $M/\overline{IO}$ ✓

1

Inv.

0

Memory — DRAM, ROM (No WE)

I/0

O_{M-1}

256k

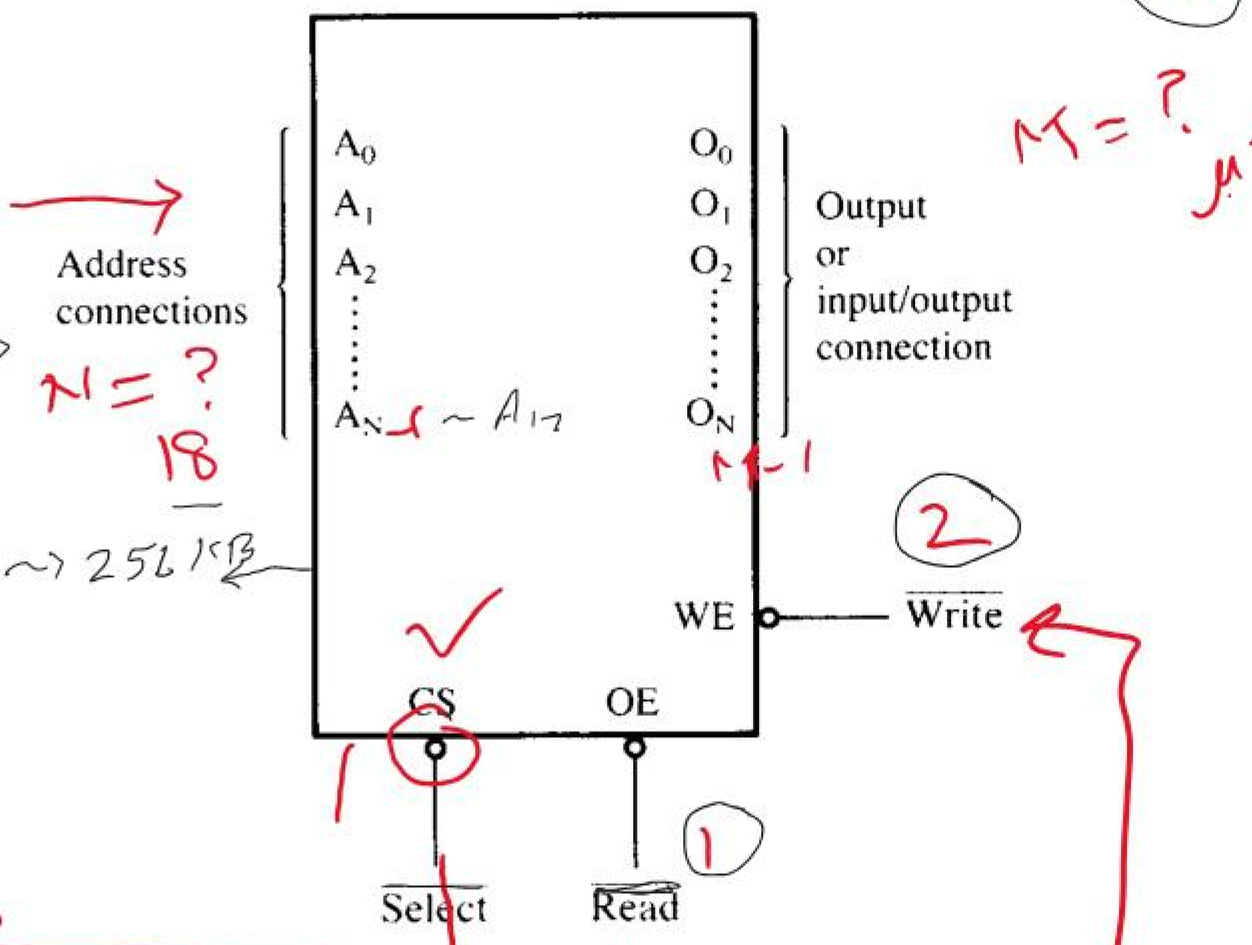
16

M = ? μD lines

N = ?

18

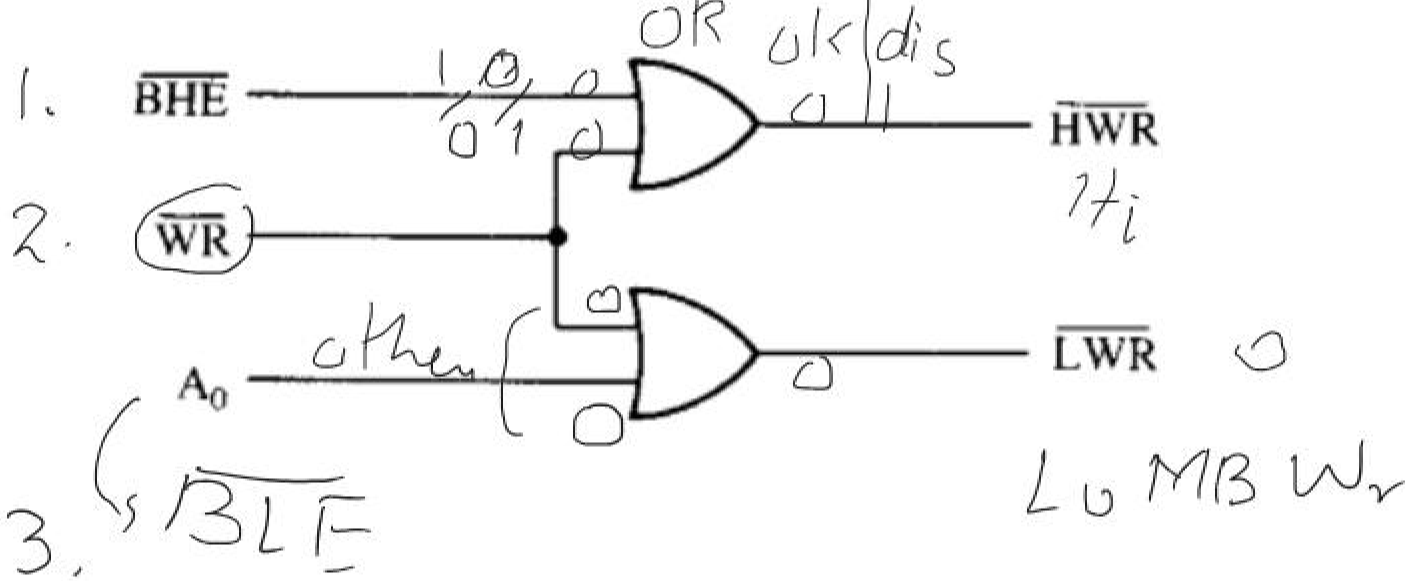
$2^{18} \sim 256KB$



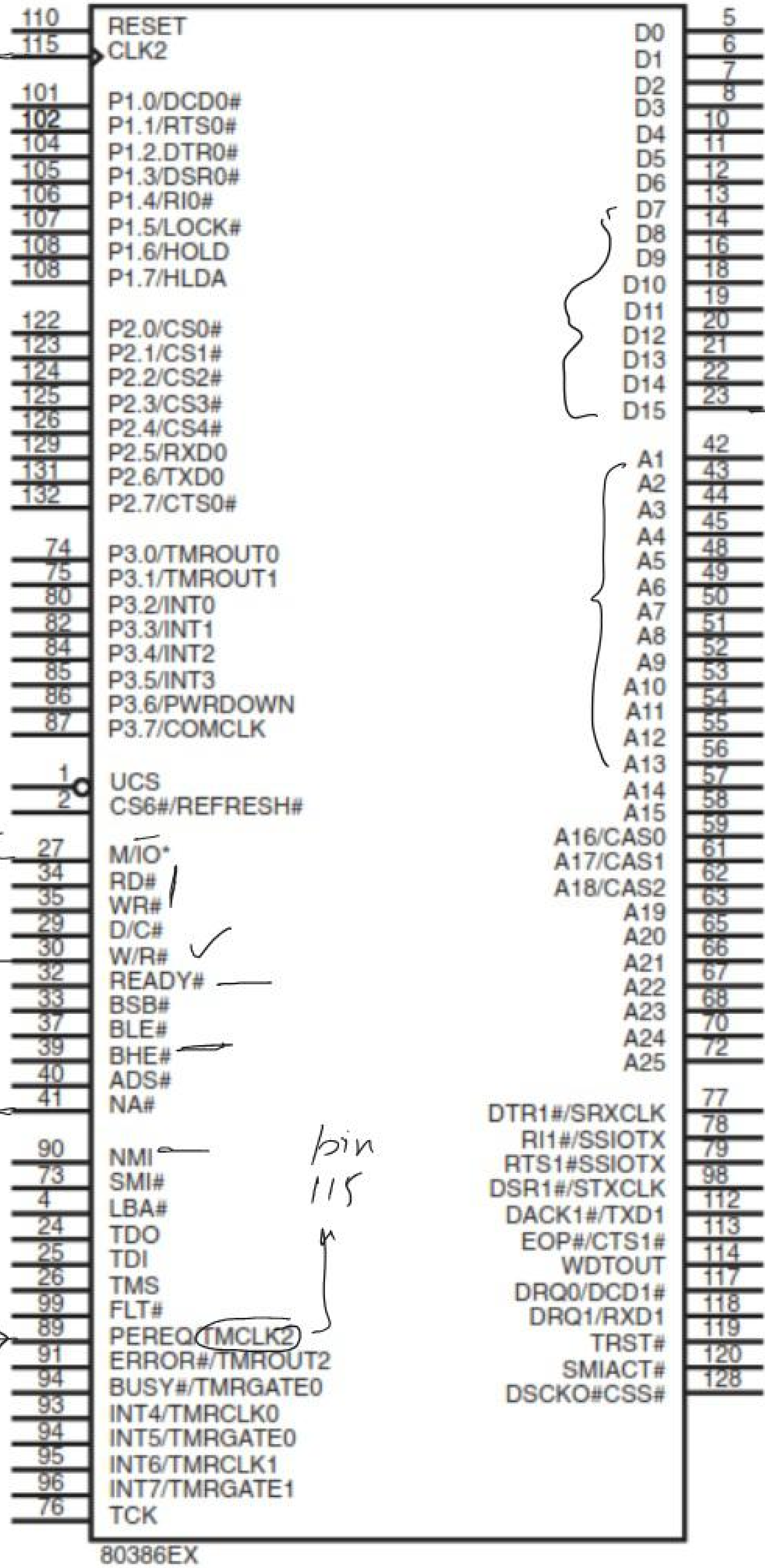
$A_0 = 0 \longrightarrow$ Address even
$A_1 = 1 \longrightarrow$ " odd $\rightarrow \overline{BLE}$ disabled
cntrl

Write

	$\overline{BHE}$	$\overline{BLE}$ / A_0	Function
1.	0	0	Both banks enabled for a 16-bit transfer
2.	0	1	High bank enabled for an 8-bit transfer
3.	1	0	Low bank enabled for an 8-bit transfer
4.	1	1	No bank enabled



- $\overline{BE_3} \sim \overline{BE_0}$: Bus enable signals for 4 Mem. Banks.
- $\overline{BS16}$: Bus select 16
  - 0 : 16-bit app
  - 1 : 32-bit app



W/R
- 0: Read
- 1: write

Pipelining * Next Address

pin 115

10MHz
↓
20MHz Clock

$D_0 \sim D_{31}$

386 DX $\sim$ 4 Bytes

D_{31}

$A_2 \sim A_{31} \sim$ 4GB

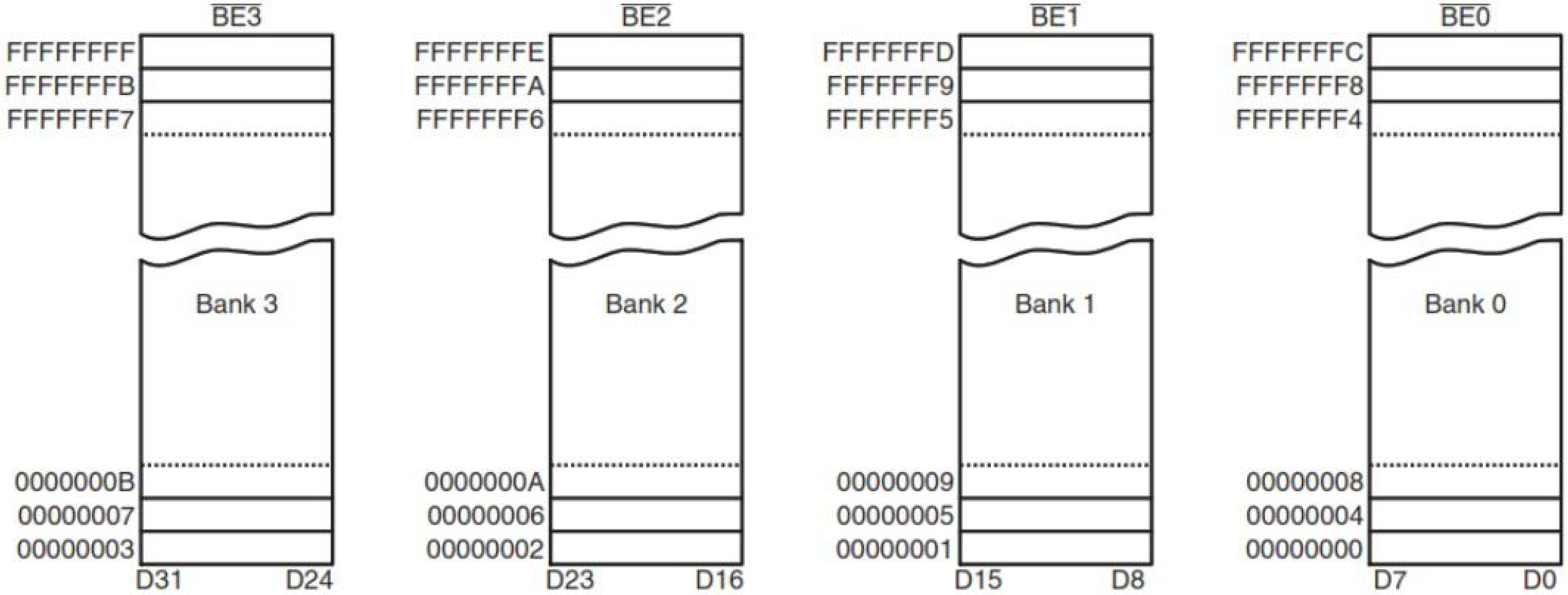


FIGURE 10–33 The memory organization for the 80386DX and 80486 microprocessors.

Internally Decoded

A_1	A_0	
0	0	— BE_0
0	1	— BE_1
1	0	— BE_2
1	1	— BE_3

FIGURE 10–34 Bank write signals for the 80386DX and 80486 microprocessors.